AI-Automated Classification of Hospital Safety Incidents

Group 888
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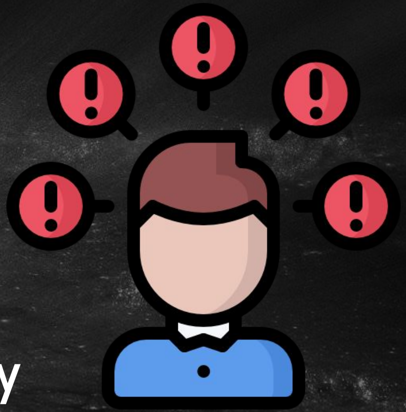
Problem



- Manual severity classification of safety incidents is:
  - Time-consuming
  - Expert-dependent
- Limits efficient, accurate monitoring of patient safety

Target Audience

- Hospital quality & safety teams
- Clinicians/nurses
- Healthcare systems
- Researchers

Solution: AI-powered, automated incident classification

Why Better?

- Review time from hours → seconds
- Instant, consistent results
- Multi-language support

Impact: free up **2,000** staff hours/year per hospital

Efficient **Scalable** **Inclusive**

Data

Brief Factual Description	HPI Label	Department	Deviation Rationale	Reach Pt Rationale	Label/Level Rationale
A patient was admitted to the critical care unit with congestive heart failure and later has a new complaint of chest pain persisting over several hours. Tylenol is administered but does not decrease the patient's pain scale rating. The Resident orders a laboratory work-up. An EKG shows that the patient is experiencing an acute ST-elevation myocardial infarction. The attending cardiologist is called, but does not respond to multiple pages. The nurse does not escalate the patient's emergent condition to other physicians or the rapid response team. The patient continues to decompensate, codes and expires.	SSE	Medicine	Yes. Although an adequate assessment indicating the patient's need for immediate intervention for acute myocardial infarction was performed, there was a significant delay in initiating treatment.	Yes	SSE 1: Death
A patient presented to the Emergency Department and is admitted to rule out myocardial infarction (M an incidental finding of a "questionable atrial throm admitting intern are notified. The ED physician inc dictated report, but the intern does not include this attending cardiologist misses this information. The an acute MI is ruled out. No assessment of the atr weeks later, the patient presents to the ED complaining of right sided flank pain. A CT scan shows a renal thrombus causing a right kidney infarction rendering the kidney non-functional which is further confirmed by additional flow scans.	SSE	Medicine	the time of the initial hospitalization, the condition was not adequately assessed or appropriately treated.	Yes	SSE 2: Severe Permanent Harm (delay in treatment resulted in loss of function of one kidney)
Patient 'A' is admitted for treatment of pneumonia. Admitting orders for "Lorazepam 1mg PO three times daily" is intended for another patient 'B' but inadvertently entered on patient 'A's' medical record. Patient A receives six doses of Lorazepam. He requires a Foley catheter and due to his confusion, pulls it out causing damage to his urethra. This required significant urological surgery to correct the urethral damage an extension of hospital stay.	SSE	Medicine	Yes. An order was written on the wrong patient. It wasn't caught by the physician, the pharmacist or the nurses administering the drug	Yes	SSE 5: Severe Temporary Harm (serious medication event leading to situation requiring significant surgical intervention and extended hospitalization)

Brief factual descriptions of the event

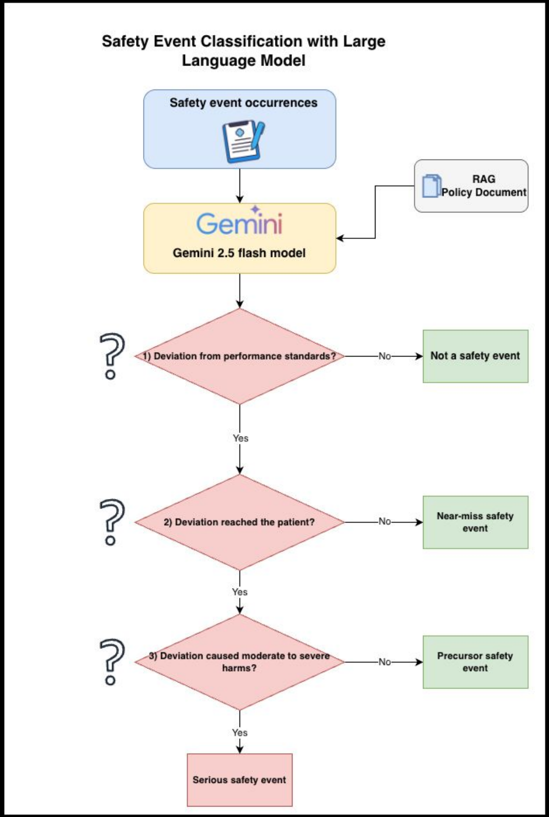
Classifications

- Not a Safety Event
- Near Miss
- Precursor Safety Event
- Severe Safety Event

Departments

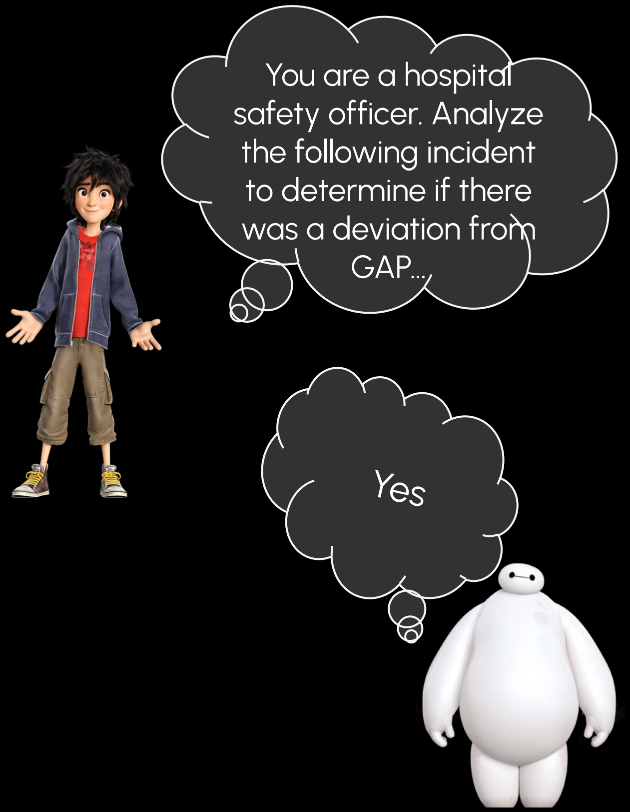
- Medicine
- Surgery
- OB/GYN/NICU
- Radiology/Imaging
- Outpatient/ER

How Our AI Thinks Like a Hospital Safety Expert



- Three precision-engineered prompts replicate the clinician's decision path
- Each prompt outputs a binary "Yes/No" similar to expert review.
- Outcome: From tedious manual review to faster, cheaper, and efficient classification

Steering AI with Smart Prompts



- Purpose-built Prompt Design: We engineered domain-specific instructions that frame the model as a hospital safety officer
- Controlled Output: Prompts constrain responses to “Yes/No” — enforcing consistent outputs
- Result: Fewer hallucinations, higher reliability, and explainable safety event classification

Scalability and Efficiency

Docker
Reproducible Environment

GCP
Elastic Cloud Infrastructure

Vertex AI
Scalable Model Hosting

- Technical Scalability
  - Supports multiple hospital departments and user roles
  - Docker + GCP enable scaling without performance loss
- Performance Optimization
  - Iterative prompt engineering
  - Integration of a RAG pipeline
  - Fine-tuning of Gemini (experimental)

Gemini
Decision Tree Reasoning

Python Pipeline
Orchestrates prompt logic and data flow

Modular Design
Supports flexible adaptation and scaling

From Concepts to Prototype

- Deployed a reproducible AI pipeline using Docker — every teammate, identical environment, zero setup friction
- Implemented prompt-based classification and rationale generation with Gemini 2.5 flash model
- Generated structured outputs on safety reports, including event categorization and short explanations


```
{
  "incident": "A patient with tuberculosis receiving isoniazid therapy under directly observ
  "gaps_deviation_check": "No",
  "reached_patient_check": "N/A",
  "harm_level_check": "N/A",
  "final_classification_code": "NSE",
  "rationale": "No deviation from Generally Accepted Performance Standards (GAPS)."
}
```


This prototype demonstrates that AI can classify safety events like human experts — in under 10 seconds.

What's Next: From Prototype to Platform

Next Steps

Complete implementing RAG

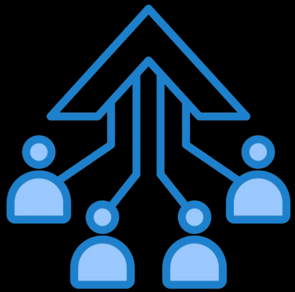
Expand input capability

Automate root cause extraction

Interactive visualization dashboard

Market Growth

- **Scalable & global**
- **Cross-hospital expansion**
- **Multilingual capability**



Thank you hehe :D